

Practical Audit Report – COMPAS Dataset Bias Analysis

The COMPAS dataset contains risk scores used to predict recidivism, with 6,150 samples after filtering for African-American and Caucasian individuals and dropping missing values. Ten features were selected for analysis, including demographic variables, prior offenses, and charge type. This dataset was converted into an AIF360 StandardDataset to facilitate fairness auditing.

Before mitigation, fairness metrics revealed significant disparities. The **Disparate Impact** in the training data was 0.811, indicating that African-American individuals were less likely to receive favorable predictions compared to Caucasian individuals. False Positive Rates (FPR) were also imbalanced: 0.318 for African-American individuals versus 0.528 for Caucasians. These differences demonstrate the model's propensity to misclassify African-American individuals as high-risk less often, while overestimating risk for Caucasians, signaling potential racial bias in predictive outcomes.

After applying a logistic regression classifier to the test set without mitigation, the **Disparate Impact** dropped further to 0.680, highlighting amplification of bias in predictions. FPRs remained unequal, confirming the ethical risk of unfair treatment in real-world decision-making contexts.

To address these issues, **Reweighting**, a pre-processing bias mitigation technique, was applied. This method adjusts instance weights to balance the impact of protected attributes during training. Post-mitigation results show a **Disparate Impact of 1.006** and nearly equal FPRs: 0.441 for African-American and 0.433 for Caucasian individuals. These values indicate that reweighting successfully reduced bias, aligning outcomes across racial groups.

Visualizations of FPRs before and after mitigation further illustrate the improvement: initial disparities are corrected, demonstrating that fairness-aware techniques can meaningfully reduce algorithmic bias.

In conclusion, this audit highlights the presence of racial bias in COMPAS predictions and shows that applying fairness interventions, like reweighting, effectively mitigates discriminatory outcomes. These findings emphasize the necessity of integrating ethical AI practices and monitoring fairness metrics in high-stakes applications.